NATURAL LANGUAGE PROCESSING (NLP)

Core concepts, text representations, models, and evaluation

1. What is NLP?

Natural Language Processing is a field of artificial intelligence concerned with enabling computers to analyse, understand, process, and generate human language. It combines linguistics, computer science, statistics, and machine learning. NLP systems commonly work with text, transcripts, or other language representations.

Applications include search, spam filtering, sentiment analysis, intent classification, machine translation, summarisation, information extraction, chatbots, and question answering.

2. Typical NLP pipeline

A pipeline converts raw language into a representation suitable for a task, applies a model, and evaluates the output.

Stage	Purpose	Example
Ingestion	Collect text and metadata	Support tickets
Cleaning	Handle encoding, markup, noise, and whitespace	Remove HTML tags
Segmentation	Split text into sentences or units	Sentence boundary detection
Tokenisation	Convert text into tokens or subwords	“playing” → “play” + “ing”
Representation	Convert tokens into numeric features	TF-IDF or embeddings
Modelling	Predict labels or generate text	Intent classifier
Evaluation	Measure performance and quality	F1-score and human review

3. Important linguistic concepts

Tokenisation: Splitting text into words, punctuation, or subword units. Transformer systems often use subword tokenisation to handle rare words and unseen combinations.

Normalisation: Making text more consistent through operations such as Unicode normalisation, lowercasing, whitespace handling, or spelling correction. Normalisation choices depend on the language and task.

Stemming and lemmatisation: Stemming heuristically removes word endings; lemmatisation maps words to dictionary base forms using linguistic information. For example, “studies” may be lemmatised to “study”.

Part-of-speech tagging: Assigns grammatical labels such as noun, verb, adjective, and adverb to tokens.

Named Entity Recognition: Detects spans referring to people, organisations, locations, dates, products, and other entity types.

Dependency parsing: Identifies grammatical relationships between words, such as subject, object, and modifier.

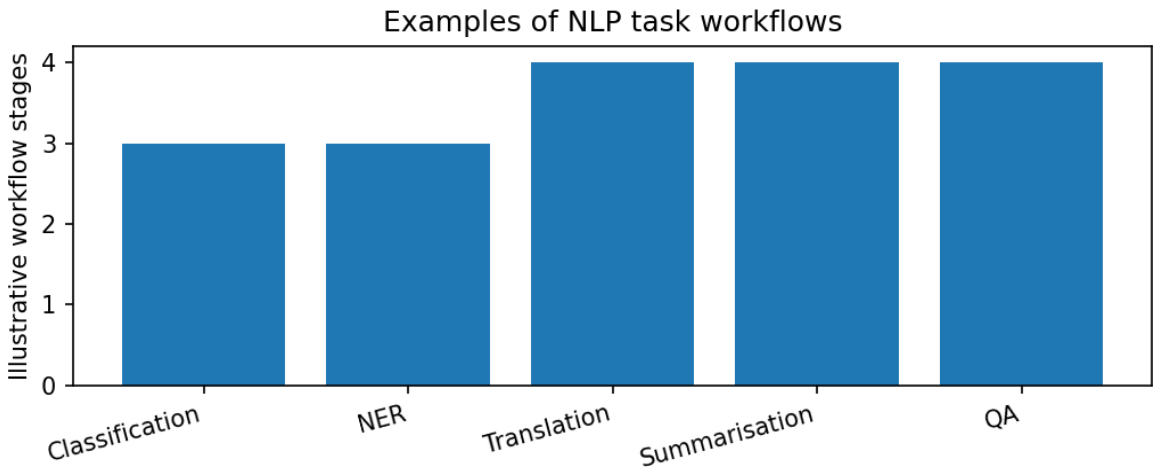
4. Text representations

Method	How it works	Strengths / limitations
Bag of Words	Represents term counts or presence	Simple; ignores word order
TF-IDF	Weights terms by document frequency and corpus rarity	Useful baseline; sparse vectors
Static embeddings	Maps each word to a dense vector	Captures similarity; same vector across contexts
Contextual embeddings	Builds representations using surrounding text	Context-aware; more computationally demanding

Embeddings are dense numerical vectors designed to encode useful semantic or syntactic information. Cosine similarity is often used to compare vectors. Embeddings support semantic search, clustering, retrieval, and downstream prediction.

5. Common NLP tasks

Task	Input	Output
Sentiment analysis	Review or message	Positive, negative, neutral
Text classification	Document or utterance	Topic, intent, or category
NER	Sentence or document	Entity spans and types
Machine translation	Source-language text	Target-language text
Summarisation	Long document	Shorter summary
Question answering	Question plus context	Answer or answer span



The chart is illustrative and does not represent a universal ranking of task difficulty.

6. Traditional NLP and transformers

Traditional NLP commonly uses n-grams, manually designed features, TF-IDF, and algorithms such as Naive Bayes, logistic regression, and support vector machines. These approaches can be efficient and interpretable for well-defined tasks.

Transformers use attention mechanisms to model relationships between tokens. Encoder models are often used for language understanding; decoder-only models are widely used for generation; encoder-decoder models are common in sequence-to-sequence tasks such as translation.

7. Evaluation metrics

Metric	Typical task	Meaning
Accuracy	Classification	Fraction of predictions that are correct
Precision	Classification / extraction	Fraction of predicted positives that are correct
Recall	Classification / extraction	Fraction of actual positives that are found
F1-score	Classification / extraction	Harmonic mean of precision and recall
BLEU	Translation	N-gram overlap with reference translations
ROUGE	Summarisation	Overlap-oriented comparison with references
Perplexity	Language modelling	How well a model predicts a token sequence

Automatic metrics do not capture every aspect of language quality. Human review may be needed for fluency, factuality, relevance, bias, safety, and task completion.

8. NLP in Retrieval-Augmented Generation (RAG)

A RAG system combines information retrieval with a language model. Documents are extracted, split into chunks, embedded, and stored in a vector index. A query is embedded and matched against relevant chunks. The retrieved text is supplied to a language model as context for answering.

• Ingest documents and extract text.

• Split content into chunks, optionally with overlap.

• Generate embeddings and store them in a vector database.

• Retrieve relevant chunks using vector or hybrid search.

• Construct context, generate an answer, and evaluate its grounding.

9. Practical considerations

• Use representative data across languages, dialects, accents, and domains.

• Check annotation consistency, duplicates, noisy labels, and class imbalance.

• Prevent data leakage between training and test sets.

• Protect personal information and apply access controls.

• Monitor changing vocabulary, user intent, and production errors.

10. Key terms

Corpus: A collection of language data. **Token:** A unit produced by tokenisation. **Vocabulary:** The set of recognised tokens. **Embedding:** A dense vector representation. **Inference:** Running a trained model on new input. **Hallucination:** Generated content that is unsupported or incorrect.

Summary

NLP turns human language into data that software can analyse or generate. Reliable NLP requires appropriate preprocessing, linguistic awareness, useful representations, suitable models, and task-specific evaluation.